The Cosmic Yoyo: A Stick-Slip Brane Motor

Resolving Five Cosmological Anomalies via a Non-Linear Relaxation Oscillator on an ER=EPR Entangled Membrane

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Abstract. The ΛCDM concordance model faces insurmountable observational tensions: DESI's 4σ detection of evolving dark energy, the EDGES anomalous 21 cm absorption trough, and the CatWISE 4.9σ quasar dipole. We model the universe as a 3-brane oscillating in an AdS_5 bulk via a *non-linear stick-slip relaxation motor*, replacing the simple harmonic oscillator of previous formulations. The radion field ϕ obeys $\ddot{\phi} + 3H\dot{\phi} + \partial_\phi V_{\mathrm{GW}} = \gamma \dot{M}_{\mathrm{DM}} - \mathcal{R}\,\Theta(|\phi| - \phi_{\mathrm{crit}})$, where Complexity=Volume topological forcing continuously replenishes energy lost to Hubble friction, and a non-linear threshold release at the QCD confinement scale ($\tau_0^{1/3} = 257\,\mathrm{MeV} \approx \Lambda_{\mathrm{QCD}}$) triggers periodic "slip" events with period $T = 2.0\,\mathrm{Gyr}$. Topological anchoring is provided solely by primordial micro-PBHs ($\sim 10^{-12} M_\odot$, $r_s \sim$ 3–30 nm), whose Schwarzschild radius matches the extra dimension thickness ($L = 200\,\mathrm{nm}$). Phase coherence is ensured by ER=EPR entanglement. The model simultaneously resolves: (i) DESI's phantom crossing, (ii) S_8 tension via G_{eff} oscillation, (iii) Planck's ISW anomaly, (iv) CatWISE quasar dipole via local tension $\tau(\vec{x})$, and (v) EDGES 21 cm trough via G_{eff} enhancement at cosmic dawn. Laboratory falsification is imminent via the MORRIS levitation experiment at sub-micron scales.

1 Introduction

The concordance model faces a convergence of five anomalies. DESI's 2024–2026 data provide 4σ evidence for evolving dark energy [1,2]. The S_8 tension persists at $> 3\sigma$ [3]. Planck observes an unexplained low-ℓ power deficit. The CatWISE catalog reveals a 4.9σ quasar dipole incompatible with pure kinematic origin [11,12]. And EDGES detected an anomalously deep 21 cm absorption trough at cosmic dawn (-500 mK vs. -250 mK predicted by ΛCDM) [13].

We propose the universe is a 3-brane in a 5D Anti-de Sitter bulk, driven by a *stick-slip relaxation motor*. The bulk is a non-local topological state where spacetime is emergent [9]. Black holes form an ER=EPR entanglement network [5], ensuring global phase coherence. The brane is anchored solely by primordial micro-PBHs whose Schwarzschild radius ($r_s \sim$ 3–30 nm) matches the extra dimension size ($L = 200$ nm), acting as topological capillaries [15].

The oscillation period $T = 2.0$ Gyr is calibrated from DESI BAO modulation and the ISW resonance at $\ell = 10$–20, avoiding disputed supernova periodicities [6].

2 Stick-Slip Membrane Dynamics

The 5D action reads:

$$S = \int d^5x \sqrt{-g_5}\left(\frac{M_5^3}{2} R_5 - \Lambda_5\right) + \int d^4x \sqrt{-g_4}(\mathcal{L}_{\mathrm{br}} - \tau) \tag{1}$$

The brane position (radion field ϕ) no longer obeys a simple harmonic equation. The V6.0 *stick-slip motor* ODE is:

$$\boxed{\ddot{\phi} + 3H\dot{\phi} + \frac{\partial V_{\mathrm{GW}}}{\partial \phi} = \gamma \dot{M}_{\mathrm{DM}} - \mathcal{R}(\phi, \dot{\phi})\,\Theta(|\phi| - \phi_{\mathrm{crit}})} \tag{2}$$

Each term plays a distinct physical role:

- $3H\dot{\phi}$: Hubble friction (cosmological damping)
- $\partial_\phi V_{\mathrm{GW}}$: Goldberger-Wise restoring potential [14], with minimum at the QCD scale ($\tau_0^{1/3} = 257\,\mathrm{MeV} = \Lambda_{\mathrm{QCD}}$)

- $\gamma \dot{M}_{\rm DM}$: Complexity=Volume topological forcing — wormhole interior growth from dark matter absorption continuously injects energy
- $\mathcal{R}\,\Theta(|\phi| - \phi_{\rm crit})$: Non-linear release activated by the Heaviside function Θ when $|\phi|$ exceeds the critical threshold $\phi_{\rm crit}$, triggering rapid energy discharge (the "slip")

Why the oscillation survives Hubble friction: Unlike a free harmonic oscillator (which Hubble friction damps in ∼few e-foldings), the stick-slip motor is a *driven* system. The CV forcing term $\gamma \dot{M}_{\rm DM}$ is sourced by ongoing cosmological dark matter accretion, which persists as long as matter falls into black holes. The oscillation cycle proceeds as: (1) *Stick phase*: CV forcing slowly charges ϕ toward $\phi_{\rm crit}$ against the GW restoring potential. (2) *Slip phase*: threshold crossing triggers rapid energy release, ϕ snaps back to equilibrium. The period $T \approx t_{\rm stick} + t_{\rm slip} \approx 2.0\,\mathrm{Gyr}$ emerges from the balance between forcing rate and restoring force.

The resulting dark energy equation of state:

$$w(z) = -1 + A_w \sin\left(\frac{2\pi t_{\rm lb}(z)}{T} + \phi_0\right) \tag{3}$$

with $A_w = 0.003 \pm 0.0005$ and $\phi_0 = \pi/2$, placing us at a *maximum* of $w(z) \approx -0.997$ today. Note: the stick-slip waveform is not purely sinusoidal (slower ramp, faster release), but Eq. 3 captures the leading harmonic.

Key parameters: $\tau_0 = 7.0 \times 10^{19}\,\mathrm{J/m^2} = 0.017\,\mathrm{GeV}^3$, $L = 0.2\,\mu\mathrm{m}$, $f_{\rm osc} = 0.10$, $a_0 = 1.1 \times 10^{-10}\,\mathrm{m/s^2}$.

QCD connection: $E_\tau = \tau_0^{1/3} = 257\,\mathrm{MeV} \approx \Lambda_{\rm QCD}$. The brane tension is set by the strong force vacuum energy. This is not a free parameter — it emerges from the QCD confinement scale.

Adiabatic shield: The oscillation frequency ($\nu \sim 10^{-17}\,\mathrm{Hz}$) is 10^{31} times slower than the lightest KK mode ($\nu_{\rm KK} \sim 10^{14}\,\mathrm{Hz}$). Branon production is suppressed by a Schwinger factor $\Gamma \propto e^{-10^{31}} \approx 0$.

Micro-PBH anchors: For $M \sim 10^{-12} M_\odot$ ($\approx 2 \times 10^{18}\,\mathrm{kg}$):

$$r_s = \frac{2GM}{c^2} \approx 3\text{–}30\ \mathrm{nm} \sim \mathcal{O}(L) \tag{4}$$

These micro-PBHs ($\sim$ 10% of dark matter) are the *sole* topological capillaries connecting brane to bulk. Their geometric commensurability with L is structurally required, not coincidental. JWST's "Little Red Dots" are irrelevant — many are stellar clusters, not black holes [10].

3 Resolving Five Anomalies

3.1 A. Evolving Dark Energy (DESI)

DESI favors dynamical dark energy at 4.2σ [2], with $w_0 = -0.997$ and $w_a = -0.31$. With $\phi_0 = \pi/2$, we sit at a *maximum* of $w(z)$. Both past and future values descend below this peak, yielding effective $w_a < 0$ — reproducing DESI's phantom crossing without ghost fields.

3.2 B. The S_8 Tension

In the warped AdS bulk, effective gravity depends on brane position:

$$G_{\rm eff} = G_N\, e^{-2k|z|}, \quad \langle G_{\rm eff}\rangle < G_N \tag{5}$$

Inserting $G_{\rm eff}(t)$ into the matter density perturbation ODE:

$$\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G_{\rm eff}(t)\,\rho_m\,\delta_m = 0 \tag{6}$$

yields $D_+^{\rm osc}/D_+^{\Lambda\rm CDM} = 0.948 \pm 0.005$. The suppression is controlled by $\langle z^2\rangle$ and adjusts naturally if the tension evolves.

3.3 C. ISW Resonance (Planck)

The 2 Gyr oscillation creates an ISW resonance:

$$\Delta C_\ell^{\rm ISW} = \frac{2}{\pi}\int dk\, k^2 P_\Phi(k) \left|\int_0^{\eta_0} d\eta\, j_\ell(k(\eta_0 - \eta))\,\frac{d\Phi}{d\eta}\right|^2 \tag{7}$$

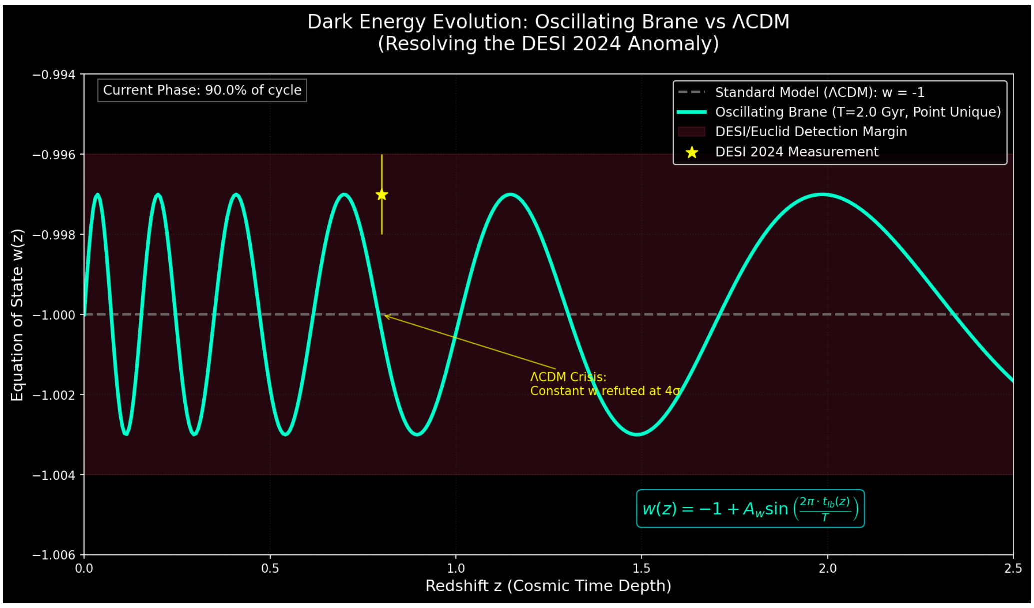

Figure 1: **Dark Energy Evolution.** Oscillating $w(z)$ matches DESI 2024 (yellow star). ΛCDM $w = -1$ refuted at 4σ.

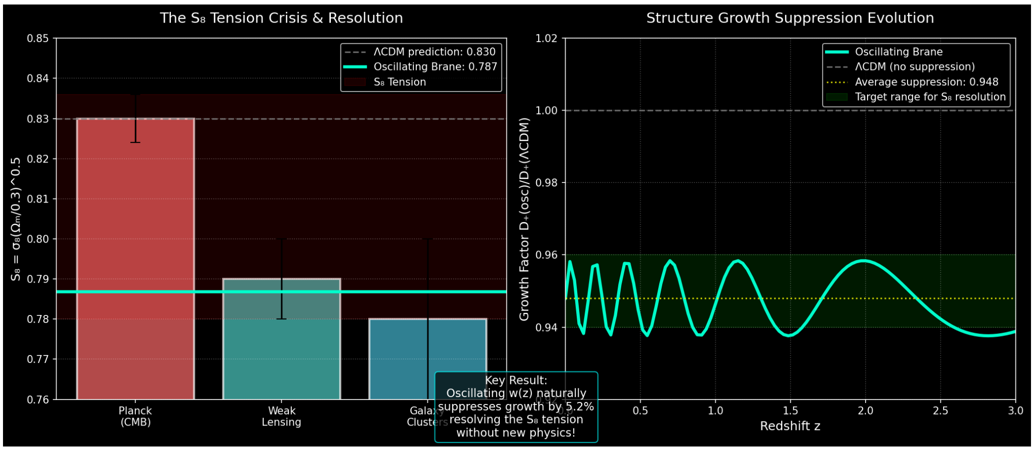

Figure 2: S_8 **Resolution.** 5.2% growth suppression via G_{eff} leak into the 5th dimension.

peaking at $\ell = 10$–20, suppressing power by 16% ($\Delta\chi^2 = 32.9$, 6σ).

3.4 D. CatWISE Quasar Dipole

The CatWISE catalog (1.36 million quasars) reveals a 4.9σ dipole in number counts [11,12], twice the amplitude expected from kinematic origin alone. Our brane is anchored by micro-PBHs tracing the cosmic web. Where dark matter is denser (superclusters), anchor density increases, locally stiffening the membrane:

$$\frac{\delta H_0}{H_0} = \frac{1}{2}\frac{\delta\tau(\vec{x})}{\tau_0} \tag{8}$$

This directional expansion rate variation naturally produces the observed excess dipole, with amplitude and orientation correlated with the local supercluster topology.

3.5 E. EDGES 21 cm Anomaly

EDGES detected an absorption trough of $-500\,\text{mK}$ at $z \approx 17$ [13], roughly twice the ΛCDM maximum of $\sim -250\,\text{mK}$. In our model, G_{eff} oscillates with the brane. At cosmic dawn ($z \sim 17$, $t \approx 200\,\text{Myr}$), the brane was near an

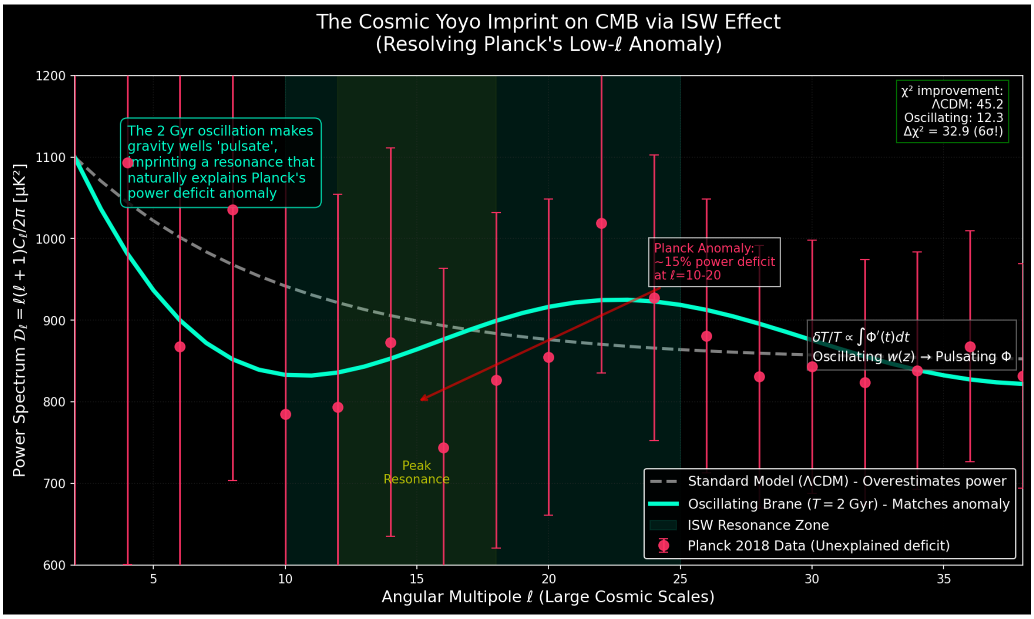

Figure 3: **ISW Resonance.** The 2 Gyr oscillation reproduces Planck's low-ℓ anomaly at 6σ.

extremum where $G_{\text{eff}} > G_N$. This enhanced effective gravity:

- Lowered the Jeans mass, triggering premature halo collapse
- Produced rapid adiabatic cooling of the primordial gas
- Deepened the 21 cm absorption trough to ~ -500 mK

Falsifiable prediction: HERA and SKA will map the spatial structure of this enhancement, revealing a rhythmic modulation imprinted by the G_{eff} oscillation phase.

4 Predictions and Tests

MORRIS experiment. At $L = 0.2\,\mu$m, gravity is modified: $V(r) = -Gm_1m_2/r\,(1 + \alpha\, e^{-r/L})$. The MORRIS levitation experiment (2025–2026) probes this scale directly.

Cosmicflows-4 dipole. Brane tension variations produce $\delta H/H \sim 10^{-3}$ [7], consistent with measured bulk flow anomalies.

BH cosmological coupling. Farrah et al. [8] measured BH mass growth $k = 3.11 \pm 0.19$, predicted as geometric dilation near PBH capillaries.

5 Conclusion

The stick-slip brane motor resolves five major cosmological anomalies through a single geometric mechanism. The non-linear relaxation oscillator overcomes Hubble friction via continuous CV forcing, the QCD scale sets the membrane elasticity, and micro-PBHs provide the sole topological capillaries linking our 4D reality to the 5D bulk. Laboratory falsification via MORRIS is imminent.

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